

Yogesh Saini

Software Engineer — Android Developer & ML

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Summary

Versatile Android Developer and ML Engineer with experience building scalable, high-performance mobile applications across the entire stack. Proficient in designing intuitive Android interfaces, robust back-end APIs, and seamless database integrations to deliver reliable, production-ready products.

Education

B.Tech in Computer Science and Engineering

Sagar Institute Of Science, Technology And Research , Bhopal

2023-2027

CGPA: 8.15 / 10

Class XII

Govt Boys H S School Rehti , Sehore

2020-2021

Score : 82.6%

Class X

Govt Boys H S School Rehti , Sehore

2022-2023

Score : 92%

Skills

Core CS Fundamentals:	Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS
Languages:	C, C++, Java, Python
Android Development:	Kotlin, MVVM Architecture, Retrofit, Coroutines, Room DB, ViewModel, LiveData, Navigation Component, Android Studio
Backend & APIs:	Flask, REST API Design, RESTful Web Services, Socket.IO, JWT Authentication, Redis, WebSockets
Database:	MongoDB, Firebase Firestore, Room DB, Redis
Machine Learning & AI:	Deep Learning, Supervised Learning, Unsupervised Learning, Scikit-Learn, Pandas, NumPy, NLP, TensorFlow, TFLite
Cloud & DevOps:	Git, GitHub, Render, Firebase (Auth, FCM, Firestore), Cloudinary

Projects

DualShield AI | Kotlin, Python, TensorFlow, Flask, REST API, Sensor Fusion

Mar 2025 – Apr 2025 | github.com/yogeshsaini7172/DualShield-AI-road-accident

- Built a production-grade Android application with a dual-detection mechanism (accelerometer + ML model) using TensorFlow to detect road accidents in real time and trigger automated emergency SOS alerts via REST API.
- Trained and deployed ML models for accident classification using Python and TensorFlow; built a Flask-powered web dashboard for live incident monitoring and alert tracking.
- Implemented sensor fusion combining accelerometer and GPS data to minimise false positives; system architecture designed for low-latency response to reduce emergency notification time.

AgriSense AI | Kotlin, Jetpack Compose, MVVM, Python, TensorFlow, Scikit-Learn, XGBoost, FAISS, LLM (Llama), Flask, REST API

Jan 2025 – May 2025 | github.com/yogeshsaini7172/Agriculture-disease-detection

- Designed and shipped a full-stack Android + AI application for real-time crop disease detection using a MobileNetV2 deep learning model via REST API, achieving 98.7% classification accuracy across 5 disease types; integrated Explainable AI (XAI) via LIME for transparent predictions.
- Built a RAG (Retrieval-Augmented Generation) pipeline with FAISS vector database and Llama LLM to deliver context-aware agronomic advice aligned with ICAR guidelines, reducing manual expert consultation dependency.
- Developed a precision fertilizer recommendation engine using Random Forest & XGBoost classifiers trained on soil NPK levels, weather API data, and crop type, improving input guidance accuracy.
- Implemented Agile-style iterative development; added multilingual support (English, Hindi, Punjabi), voice TTS/STT chatbot, IoT sensor integration, and a farm performance analytics dashboard.

Digital Pass | *Kotlin, Jetpack Compose, MVVM, Python, Flask, REST API, MongoDB, Firebase (Auth, FCM), Socket.IO, Redis, Room DB, Retrofit, Coroutines, JWT, Google Cloud*

Sep 2025 – Jun 2026 | github.com/yogeshsaini7172/digitalPassCollabrativeRepository

- Architected a multi-role campus security platform serving 7 user roles (Admin, Principal, HOD, Faculty, Student, Security Guard, Reception) with role-based access control (RBAC), JWT authentication, and Firebase Cloud Messaging (FCM) push notifications.
- Designed a scalable RESTful Flask backend with MongoDB Atlas for persistent storage, Redis caching for performance optimisation, and Flask-SocketIO / WebSockets for real-time gate pass approvals with sub-second latency.
- Implemented clean MVVM architecture on Android with Room DB for offline-first data persistence, async REST API calls via Retrofit & Kotlin Coroutines, GPS location services via OpenStreetMap, and Cloudinary for image management.
- Led collaborative development using Git/GitHub branching strategy; delivered the project end-to-end from system design to deployment on Render cloud platform.

Certifications

- Introduction to Cybersecurity — Cisco Networking Academy
- Python Essentials 1 — Cisco Networking Academy
- Python Essentials 2 — Cisco Networking Academy
- CCNA : Introduction to Networks
- CCNA : Enterprise Networking, Security, and Automation
- CCNA : Switching, Routing, and Wireless Essentials

Achievements

- Solved **300+** DSA problems across LeetCode, GeeksforGeeks and Coding Ninjas demonstrating advanced algorithmic problem-solving skills.